

Phase 1: Data Collection

INPUT

- APNIC extended statistics
- RIPE Stats API
- VNIX website
- BGP RIB dump

PROCESS

- Extract baseline 'VN' ASNs & prefixes
- Scrape VNIX peering members
- Parse BGP dumps & extract AS_PATHs

OUTPUT

- List of all ASNs in Vietnam
- Prefixes for all Vietnam ASNs
 - Raw AS paths

Phase 2: Data Filtering and Enrichment

INPUT

- PeeringDB
- CAIDA dataset (3 sets)
- IANA special registry

PROCESS

- Cross-filter 'VN' code & resolve WHOIS
- Filter private/reserved & Route Servers
- Map entities & classify edges (p2p/p2c)

OUTPUT

- Facilities, IXPs, IXP networks
- Cleaned & classified AS edges

Phase 3: Graph Construction and Topological Analysis

INPUT

- Processed AS Nodes & Edges
- Infrastructure Data

PROCESS

- Mathematical modeling (NetworkX)
- Calculate metrics (centrality, etc.)
- Render graph layout (ForceAtlas2)

OUTPUT

- Macroscopic AS-level graph
- Topological stats and metrics

Feeds into

Feeds into